

Roll No.

TCS-101

**B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

**FUNDAMENTAL OF COMPUTER AND INTRODUCTION TO
PROGRAMMING**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question except question No. 2.

(iii) Attempt all parts of question no. 2.

(iv) Total marks in each main question are **twenty**.

(v) Each sub-question carries 10 marks.

1. (a) What is software ? Explain various types of software. In which category Operating System falls ? Also describe the various features of OS.
- (b) Draw a flowchart to input angles of a triangle and check the validity of the triangle. (no angle should be zero).
- (c) Draw a flowchart to input n different positive integers and print factors of each number.

P. T. O.

2. (a) Assuming no syntax error in the following segments of code, what should be the output of the following ?

(i) #include <stdio.h>

```
int main()
```

```
{
```

```
    int a=2,b=3,c=0;
```

```
    if(a=a^b)
```

```
    {
```

```
        while(!(b=a^b))
```

```
        {
```

```
            c=a^b;
```

```
        }
```

```
        a=a^b;
```

```
    }
```

```
    else
```

```
    {
```

```
        b=a^b;
```

```
    }
```

```
    printf("%d %d %d",a,b,c);
```

```
    return 0;
```

```
}
```


(ii) #include <stdio.h>

```
int p(int i,int n)
{
    if(i==1)
        return 1;
    if(!(n%i))
        return 0;
    return p(i-1,n);
}
int main()
{
    if(!p(6,7))
        printf("NO");
    else
        printf("Yes");
    return 0;
}
```

(iii) #include <stdio.h>

```
int main()
{
    int a=2,b=3,c=0, n;
    c=b<<a;
    n=a>>1;
    a=a | b;
    printf("%d %d %d", a,b,c);
    return 0;
}
```


(iv) #include <stdio.h>

```
int main()
{
    int i, j;
    for(i=0; i<=5; i++)
    {
        for(j=0; j<=i; j++)
            printf("#");
        printf("\n");
    }
    return 0;
}
```

(v) #include <stdio.h>

```
int main()
{
    printf("Hello\rbye\n");
    printf("Graphic\tEra\vDeemed\nto
be\nrUniversity\n");
    return 0;
}
```

(vi) #include <stdio.h>

```
int main()
{
    unsigned static int a=65;
    printf("%d,%c", a, a);
    return 0;
}
```



```
(vii) #include <stdio.h>
int main()
{
    int a=5;
    switch(a)
    {
        case 1: printf("1\n");
        case 2: printf("2\t");
        case 3: printf("3\n");
        case 4: printf("4\t");
        case 5: printf("5\n");
        case 6: printf("6\t");
        default: printf("default"); break;
    }
    return 0;
}
```

```
(viii) #include <stdio.h>
int main()
{
    int a=1,b=3,i=0;
    while(i<1)
    {
        printf("%d,%d\n",a,b);
        a=a+b;
        b=a-b;
        a=a-b;
        i++;
    }
    printf("%d,%d\n",a,b);
    return 0;
}
```


(ix) #include <stdio.h>

int main()

{

int a=1,b=2,c,i;

printf(“%d %d %d %d”,0,1,1,2);

for(i=5;i<10; i++)

{

printf(“%d”,a+b);

c=b;

b=a+b;

a=c;

}

return 0;

}

(x) #include <stdio.h>

int main()

{

int a=5,n=0,i;

for(i=1;i<=a;i++)

n*=i;

printf(“1×2×...×a is : %d”,n);

return 0;

}

- (b) Describe life cycle of 'C' program. Explain various steps which are involved for converting a source code to executable code.

3. (a) Explain the various decision-making and looping construct available in 'C'.
- (b) There exist infinite tiles numbered 1, 2, 3, 4, 5, and so on and a hungry monkey. Monkey who is already sitting on tile numbered 1, eat one banana and decides to jump on tile numbered 4 by skipping 2 tiles and eat four bananas. After eating, it then decides to jump to tile numbered 9 by skipping 4 tiles and eat nine bananas. Then, it decides to jump to tile numbered 16 by skipping 6 tiles and eat 16 bananas.
- Write a C program to check how many bananas monkey will eat if 'n' number tile is entered as input :

Sample Input (tile number)	Sample Output (number of bananas)
1—3	1
4—8	4
9—15	9
16—24	16
and so on	

- (c) Manoj is learning programming these days. His teacher decides to teach him a new topic through which Manoj would be able to input a line number 'n' and print pyramids of various kinds. While revising the classwork, Manoj thought to try on a new pyramid of numbers ranging 1 to 9.

Manoj writes the pattern on paper, which is :

```

1
2 3
4 5 6
7 8 9 1
2 3 4 5 6
7 8 9 1 2 3
.... and so on till nth line.
```


Even though, Manoj tries hard but is not able to reproduce this pattern in which he failed and is not able to understand the solution. Write a C program to help Manoj in the task.

4. (a) Explain various storage classes available in 'C' with suitable example.
 - (b) Write a 'C' code to pass a positive integer number into a function named strong() and check whether number is strong number or not. Use a separate function fact() to find the factorial of number. (Strong number is a number whose sum of factorial of each digit is equivalent to the original number. Ex. 145)
 - (c) What is recursion ? How is it useful in programming ? Write a recursive code to find sum of digit of a number.
5. (a) When is it useful to use array ? Describe the merits and demerits of the array. Also explain the initialization methods of array.
 - (b) Mr. Anil teaches in a class of one hundred students. He decides to shortlist top ten students from the class for participating in a quiz competition, to be held in national capital the following month. For this purpose, he conducts a test, and selects top ten highest marks scoring students. Write a C program to input marks of the students and print marks of those students who are shortlisted by Mr Anil.
 - (c) Write a 'C' code to input n integer numbers in 1D array. Double each even number and triple each odd number. Print the final array.

Roll No.

TEE-101

B. TECH. (FIRST SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

BASIC ELECTRICAL ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the following terms :

(CO1, CO2, CO3)

(i) Ideal independent voltage source and practical voltage source.

(ii) Active and passive circuit elements.

(iii) Cycle and time period in single phase AC circuit.

(iv) Linear and non-linear circuit elements.

(v) Current division rule in DC circuit.

P. T. O.

- (b) Use nodal analysis to find the currents in various resistors of the circuit as shown in Fig. 1. (CO1, CO2, CO3)

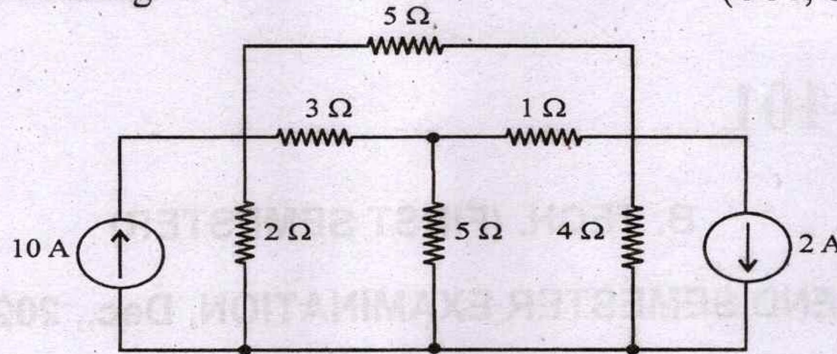


Fig. 1

- (c) The instantaneous values of two alternating voltages are represented by $v_1 = 60 \sin \theta$ and $v_2 = 40 \sin \left(\theta - \frac{\pi}{3} \right)$. Derive the expression for the instantaneous values of (i) the sum and (ii) the difference of these voltages. (CO1, CO2, CO3)

2. (a) State Thevenin's theorem for DC circuits and also draw the Thevenin's equivalent circuit. (CO1, CO2, CO3, CO4)

Determine the current in the 5Ω resistor connected across the terminals A-B of the circuit shown in Fig. 2 using Thevenin's theorem.

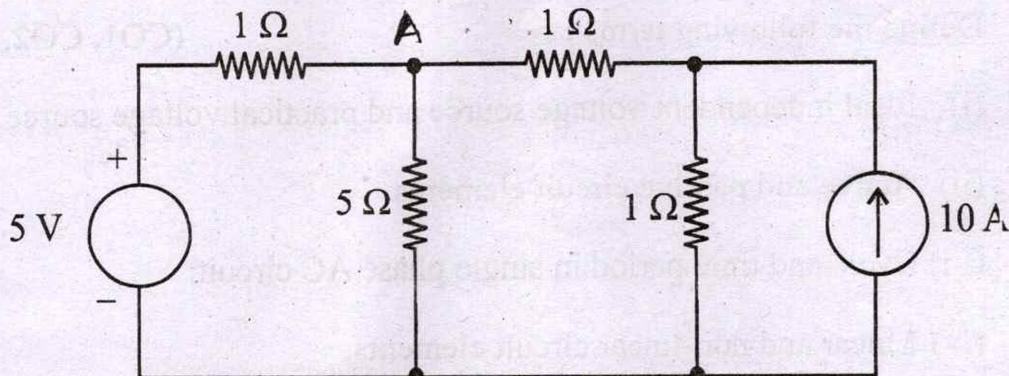


Fig. 2

- (b) Explain the working of single phase purely capacitive circuit with proper phasor diagram between the supply voltage and current. Also derive the expression for the instantaneous power and average power.

(CO1, CO2, CO3, CO4)

- (c) What do you understand by Circuit Breakers ? Give classification of circuit breakers. What is the difference between miniature circuit breaker (MCB) and moulded-miniature circuit breaker (MCCB) ?

(CO1, CO2, CO3, CO4)

3. (a) For a single phase series R-C a.c. circuit define the following terms with the phasor diagram :

(CO1, CO2, CO3, CO4)

- (i) Voltage triangle
- (ii) Impedance triangle
- (iii) Power triangle

- (b) State Superposition theorem for DC circuit.

Using star-delta transformation determine the resistance between terminals $a-b$ and the total power drawn from the supply in the circuit shown in Fig. 3.

(CO1, CO2, CO3, CO4)

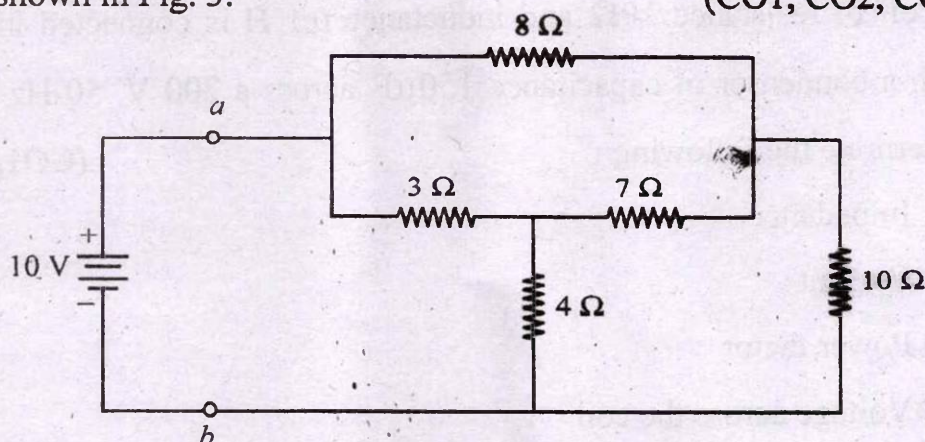


Fig. 3

- (c) Define the term "Electrical Cable". Give classification of electrical cable. Explain any two in detail. (CO1, CO2, CO3, CO4)
4. (a) A resistance and an inductance are connected in series across a voltage $v = 283 \sin(314 t)$. The current expression is found to be $i = 4 \sin(314t - \pi/4)$. Find the values of resistance, inductance and power factor. (CO1, CO2, CO3)
- (b) State Maximum Power Transfer theorem for DC circuit. Derive the condition of maximum power transfer in DC circuit. Also derive the expression of maximum power and the maximum efficiency in DC circuit. (CO1, CO2, CO3)
- (c) Define the following terms : (CO1, CO2, CO3)
- (i) KVL
 - (ii) Root mean square value
 - (iii) Phasor diagram
 - (iv) Open circuit condition
 - (v) Mesh current analysis
5. (a) A coil of resistance 10Ω and inductance 0.1 H is connected in series with a condenser of capacitance $150\mu\text{F}$ across a 200 V , 50 Hz supply. Determine the following : (CO1, CO5)
- (i) Impedance
 - (ii) Current
 - (iii) Power factor
 - (iv) Voltage across the coil
 - (v) Voltage across the condenser

(b) Calculate the following in the polar form :

(CO1, CO5)

(i) $(15 + j20) * (20 + j30)$ (Multiplication)

(ii) $(6 + j7) / (5 - j3)$ (Division)

(c) What do you understand by “Electrical Grounding” in an electrical system ? What are the basic needs of electrical earthing (grounding) ? Also define different types of electrical earthing. (CO1, CO5)

(COI: CO2)

(b) Calculate the following in the polar form.

$$(i) (15 + j20) * (30 + j30) \text{ (Multiplication)}$$

$$(ii) (6 + j7) / (2 - j3) \text{ (Division)}$$

(c) What do you understand by "Electrical Grounding" in an electrical system? What are the basic needs of electrical earthing (grounding)?
 Also define different types of electrical earthing.
 (CO1: CO2)

Roll No.

TEC-101

**B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

BASIC ELECTRONICS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Solve the following K-Map and realize the simplified equation using NAND Gate : (CO1)

$$F(A, B, C, D) = \sum m(0, 2, 3, 4, 6, 10, 11) + \sum d(12, 14)$$

- (b) Find the Base ($x = ?$, $y = ?$) : (CO1)

$$(40)_x = (30)_y$$

$$(35)_x = (27)_y$$

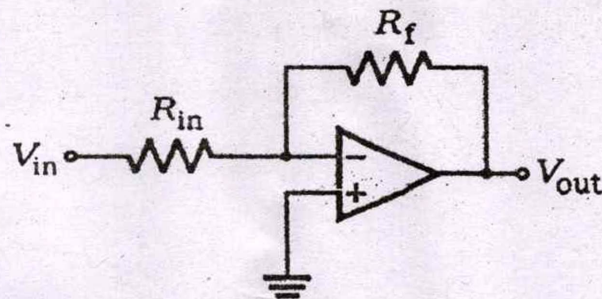
- (c) What do you understand by basic and universal gate ? List all the basic and universal gates with their symbols and truth table. Realize XOR gate using NAND Gates only. (CO1)

P. T. O.

2. (a) What is drift and diffusion current ? Derive an expression for drift and diffusion current. (CO2)
- (b) With the help of relevant diagrams explain the forward and reverse bias condition of P-N junction diode. Also, draw the V-I characteristics for the same. (CO2)
- (c) Calculate the current produced in a Ge plate of area 1.5 cm^2 and thickness 0.4 mm when a potential difference of 2 V is applied across the faces. Given that the concentration of free electrons is $2 \times 10^{19}/\text{m}^3$ and mobility of electrons and holes are $0.25 \text{ m}^2/\text{V-sec}$ and $0.07 \text{ m}^2/\text{V-sec}$ respectively. (CO2)
3. (a) Explain the working of full wave bridge rectifier with appropriate circuit diagram as well as input/output waveforms. Derive the rms current for full wave rectifier as well. (CO3)
- (b) 220 V , 50 Hz voltage is applied to the primary of a $10 : 1$ step down center tapped transformer used in full wave rectifier having load of $1 \text{ k}\Omega$. If diode resistance is 80Ω , determine : (CO3)
- (i) I_m, I_{dc}, I_{rms}
 - (ii) ac power input
 - (iii) dc power output
 - (iv) efficiency of rectification
 - (v) output frequency
- (c) Derive the ripple factor (γ) for half wave as well as full wave rectifier. (CO3)

(3)

4. (a) (i) Derive the relationship between I_{CEO} and I_{CBO} . In terms of leakage current, which configuration is better between common emitter and common base and why ?
- (ii) In a transistor if $\alpha = 0.995$ and emitter current is 5 mA, I_{CBO} is $0.25 \mu\text{A}$, then determine I_C , I_B , β and I_{CEO} . (CO4)
- (b) Draw the common base configuration circuit and explain its current gain, input and output characteristics. (CO4)
- (c) Explain the working of BJT with its operating modes. Compare the CB and CE configurations on the basis of Input impedance, Output impedance, Current gain and Voltage gain. (CO4)
5. (a) (i) Derive an expression for the output voltage of summing amplifier.
- (ii) Determine the value of output voltage V_{out} and feedback resistor R_f , such that the close loop gain is 10, $V_{in} = 5 \text{ V}$ and $R_1 = 10 \text{ k}\Omega$, in the circuit given below. (CO5/CO6)



- (b) Draw and explain the inverting and non-inverting amplifier circuit. Also, derive an expression for their gain as well. (CO5/CO6)
- (c) Derive a formula for the output voltage of difference amplifier along with its circuit diagram. (CO5/CO6)

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1:30 to 2:30



Roll No.

TCE-101

**B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

BASIC CIVIL ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) List down *five* different types of civil engineering structures and their utilization to serve human establishment and society at large. (CO1)
- (b) Discuss the following branches of Civil Engineering in terms of their role and responsibilities : (CO1)
 - (i) Geomatic Engineering
 - (ii) Environmental Engineering
 - (iii) Transportation Engineering
 - (iv) Water Resource Engineering
 - (v) Structural Engineering

P. T. O.

- (c) We all know that the infrastructure plays an important role in the economic development of a nation, so can you list down 10 points to prove the above statement. (CO1)

2. (a) In traversing in clockwise direction, the following readings were observed :

Line	Fore Bearing
AB	60°30'
BC	122°0'
CD	46°0'
DE	205°30'
EA	300°0'

Draw a neat sketch of the traverse. Determine the interior angles of the traverse and apply check. (CO2)

- (b) What is purpose of ranging in linear measurement and how it can be done ? (CO2)

- (c) Discuss the classification of surveying based on purpose of survey. (CO2)

3. (a) Explain the term Mortar. Classify it based on the type of binder used. (CO3)

- (b) An oven dried brick was weighed as 3.6 kg. It was immersed in water for 24 hours and after that saturated surface dried brick weighed 3.89 kg. Calculate the water absorption of the brick. If the acceptable limit is 5% whether the brick will be accepted for the construction purpose ? (CO3)

(3)

- (c) Discuss the following types of cement : (CO3)
- (i) Ordinary Portland cement
 - (ii) Rapid Hardening Portland cement
 - (iii) High Alumina Cement
 - (iv) Portland Pozzolana Cement
 - (v) Portland Slag Cement
4. (a) Briefly discuss the following types of buildings as per NBC : (CO4)
- (i) Educational Building
 - (ii) Institutional Building
 - (iii) Storage Building
 - (iv) Residential Building
 - (v) Assembly Building
- (b) Enlist the functions of foundation. Discuss the criteria for selection of suitable foundation. (CO4)
- (c) Discuss about the orientation of a building with the help of reference diagram. (CO4)
5. (a) In the waste water treatment, discuss the (i) Primary treatment (ii) Secondary treatment (iii) Tertiary treatment. (CO5)
- (b) Enlist some applications of transportation engineering around you and how it is useful in your daily routine ? (CO5)
- (c) Write down some water quality parameters along with the standard values assigned to them. If not what will be its impact on human ?(CO5)

(CO1)

(c) List the following types of cement:

- (i) Ordinary Portland Cement
- (ii) Rapid Hardening Portland Cement
- (iii) High Alumina Cement
- (iv) Portland Pozzolana Cement
- (v) Portland Slag Cement

(CO1)

(d) Briefly discuss the following types of buildings as per NBC:

- (i) Educational Building
- (ii) Institutional Building
- (iii) Storage Building
- (iv) Residential Building
- (v) Assembly Building

(e) Explain the functions of foundation. Discuss the criteria for selection of suitable foundation.

(CO1)

(f) Discuss about the orientation of a building with the help of reference diagram.

(CO1)

3. (a) In the waste water treatment discuss the (i) primary treatment

(CO1)

(ii) secondary treatment (iii) tertiary treatment

(b) Explain some applications of disinfection in engineering around you and

(CO1)

how it is carried out in daily routine.

(c) Write down some water quality parameters along with the standard

values assigned to them. If not what will be its impact on human? (CO1)

(CO1)

Roll No.

TBT-101

B. TECH. (BIOTECHNOLOGY) (FIRST SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

FUNDAMENTALS OF MEDICAL BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Comment on the general morphology, source of infection of *Clostridium tetani*. Differentiate the types of tetanus caused by *C. tetani*. (CO1)
- (b) Comment on the natural microflora of skin. Also, discuss the significant properties of skin which protects humans against pathogenic microbial infection. (CO1)
- (c) Using specific examples of each, classify the bacteria on the basis of capsule, endospores and flagella. (CO1)

P. T. O.

2. (a) Discuss the characteristics of viruses. How can we cultivate viruses using hen's egg embryo ? (CO2)
- (b) Describe the morphology of Rabies virus. Using diagram, explain the pathogenesis and laboratory diagnosis of Rabies infection. (CO2)
- (c) Comment on the general characteristics of fungi. What are dimorphic fungi ? Discuss the methodology followed for growth and microscopic examination of *Candida* species. (CO2)
3. (a) Differentiate between innate and acquired immunity. What are the components of innate immune response ? (CO3)
- (b) Using suitable diagrams, comment on the different cells of immune response. (CO3)
- (c) What are immunoglobulins ? Comment on different classes of immunoglobulins and their significance in immune response. (CO3)
4. (a) What is differential agar ? Describe the principle of MacConkey and Eosin Methylene Blue Agar used commonly for growth and differentiation of bacteria in laboratories. (CO4)
- (b) What are antibiotics ? Why Muller Hinton Agar is used specifically for the antibiotic susceptibility tests ? Using a flowchart only, discuss the steps of Kirby Bauer method of antibiotic susceptibility test. (CO4)
- (c) What is the National Program for control of vector borne infectious diseases. Comment on the different methods of quality control used in diagnostics microbiology. (CO4)

(3)

5. (a) What are vaccines ? Discuss on the types of vaccines available for the control of COVID-19 infection. (CO5)

(b) Discuss the role of genetic engineering and basic biomedical tools used in the treatment of diabetes. (CO5)

(c) What are monoclonal antibodies ? Describe the significance of Hybridoma technology in the production of monoclonal antibodies.

(CO5)

(a) What are vaccines? Discuss the types of vaccines available for the

control of COVID-19 infection. (CO2)

(b) Discuss the role of genetic engineering and basic biotechnological tools used

in the treatment of diseases. (CO2)

(c) What are monoclonal antibodies? Describe the significance of

Hybridoma technology in the production of monoclonal antibodies.

(CO2)

Roll No.

EEC-101

B. TECH. (BIOTECH.) (FIRST SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the principle of operation of transformer. Also derive the e.m.f. equation of the transformer. (CO3, CO4)
- (b) Define the following terms : (CO3, CO4)
 - (i) Magnetomotive force
 - (ii) Ampere turns
 - (iii) Reluctance

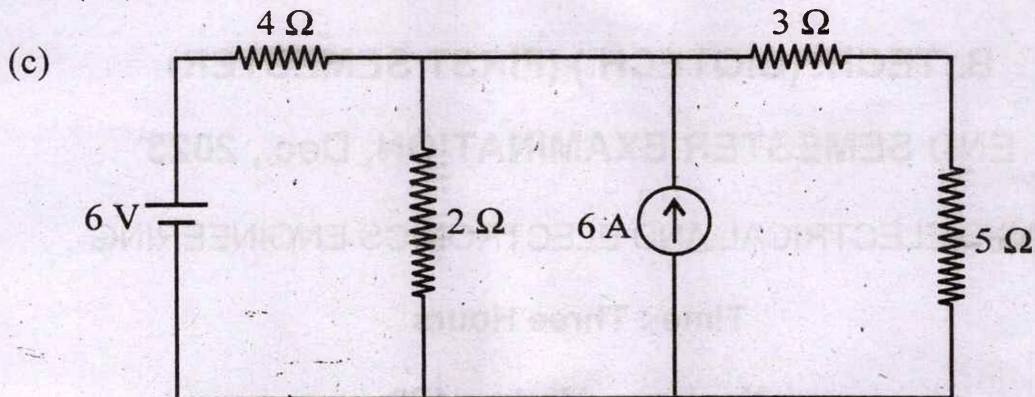
P. T. O.

(iv) Permeance

(v) Reluctivity

(vi) Eddy current loss

(vii) Core loss



- Find out the voltage drop in 5 ohms using Thevenin theorem.

(CO3, CO4)

2. (a) Discuss the similarities and differences between the electric and magnetic circuits. (CO1, CO3)

- (b) Define the following terms : (CO1, CO3)

(i) Real power

(ii) Apparent power

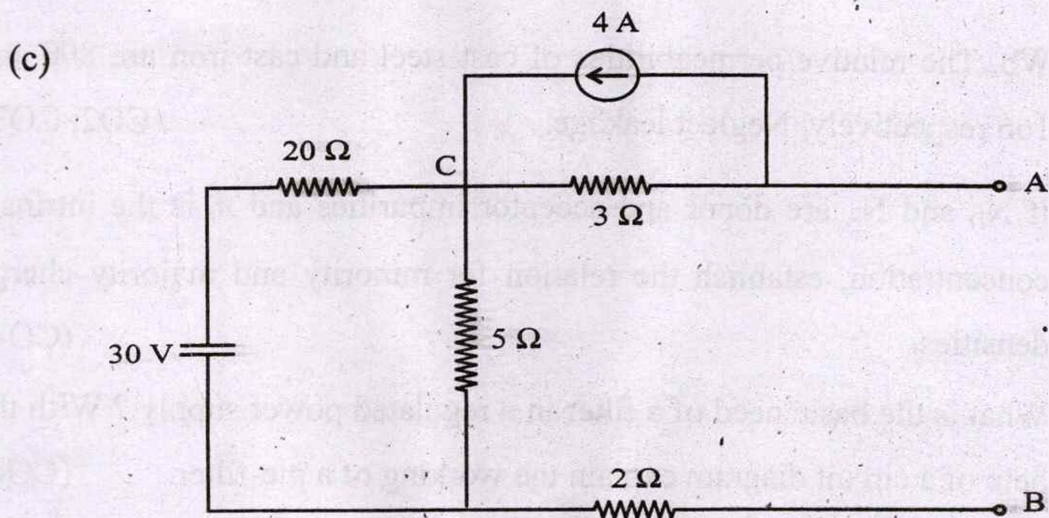
(iii) Reactive power

(iv) Power factor

(v) Inductive reactance

(vi) Capacitive reactance

Also specify the units of measurement for all the above mentioned terms.



Find out the value of the resistance to be connected across A-B for the source to supply maximum power to the load. Also find the maximum power. (CO1, CO3)

3. (a) Discuss the transformer and its types in terms of voltage output.

A single phase transformer has 600 turns on the primary and 80 turns on the secondary. If the primary is connected to a 3300 volt supply, find the secondary voltage. If the secondary current is 200A, what is the primary current? (CO2, CO3)

- (b) Explain the R-L circuit with ac supply. Derive the expressions for current and instantaneous power. Draw the phasor diagram for the same.

(CO2, CO3)

- (c) A ring has a mean diameter of 21 cm and a cross sectional area of 10 cm^2 . The ring is made up of semicircular sections of cast iron and cast steel, with each joint having a reluctance equal to an air gap of 0.2 mm. Find the ampere turns required to produce a flux of 8×10^{-4}

Wb. The relative permeabilities of cast steel and cast iron are 800 and 166 respectively. Neglect leakage. (CO2, CO3)

4. (a) If N_D and N_A are donor and acceptor impurities and n_i is the intrinsic concentration, establish the relation for minority and majority charge densities. (CO4)
- (b) What is the basic need of a filter in a regulated power supply ? With the help of a circuit diagram explain the working of a pie-filter. (CO4)
- (c) Compare the rectification efficiency of Half wave rectifier with that of the full wave rectifier. (CO4)
5. (a) Explain the working principle of N-P-N transistor with the help of a suitable diagram. (CO5, CO6)
- (b) Define the DC current gains, ' α ' and ' β ' and ' γ ' of a transistor and derive the relation between ' α ' and ' γ '. (CO5, CO6)
- (c) The reverse leakage current of the transistor when connected in CB configuration is $0.2 \mu A$ while it is $18 \mu A$ when the same transistor is connected in CE configuration. Calculate " α DC" and " β DC" of the transistor. (CO5, CO6)

Roll No.

TPH--101

B. TECH. (FIRST SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

ENGINEERING PHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Find an expression for the intensity of two waves with the same amplitude at a point in the interference region. Discuss the relationship between phase and path difference of the two waves. (CO1)
- (b) Give the theory of plane transmission grating and show how you would determine the wavelength of light. {Derive $(e + d) \sin \theta = n\lambda$ }. (CO1)
- (c) How many orders will be visible if the wavelength of the incident radiation is $6000 \text{ \AA}$ and the number of lines of the grating is 2700 in one inch ? Calculate the maximum resolving power of the grating. (CO1)

P. T. O.

2. (a) What is the polarized light ? How will you produce and detect plane, elliptically, and circularly polarized light (with a schematic diagram) ?

(CO2)

- (b) Describe the three-level laser action of the ruby laser and discuss also design and operation using the required diagram. (CO2)

- (c) A 5% solution of cane sugar placed in tube of length 40 cm, causes the optical rotation of 25° . How much length of 10% solution of the same substance will cause 45° rotation ? (CO2)

3. (a) Write the two postulates of the special theory of relativity and derive the Lorentz transformation equations. (CO3)

- (b) Derive the mass-energy relation. The mass of a moving electron is 21 times its rest mass. Calculate its kinetic energy and momentum. (CO3)

- (c) Show that $x^2 + y^2 + z^2 - c^2 t^2$ is invariant under Lorentz transformation. (CO3)

4. (a) Show the of de-Broglie wavelength in various forms (energy, voltage and temperature). Calculate the de-Broglie wavelength of (i) a particle accelerated by a potential difference of 40 kV (ii) an electron moving with a velocity of $0.2c$, where c is the velocity of light. (CO4, CO5)

- (b) Compute the electron wave function, probability density function, and energy of an electron inside the one-dimensional infinite well.

(CO4, CO5)

- (c) Discuss the quantum superposition and quantum entanglement concepts. Also, discuss the challenges that quantum hardware faces. (CO4, CO5)

(3)

5. (a) Discuss the phenomenon of superconductivity and its type. Show that in superconducting state current can flow in the superconducting specimen even if the Electric field(E) is removed (i.e. persistent current).

(CO5, CO6)

- (b) Write all four Maxwell's equations in differential form. Derive the first, second and third Maxwell's equations and discuss their significance.

(CO5, CO6)

- (c) Discuss the change in band gap if the particle size is reduced to nanometer.

(CO5, CO6)

(a) Discuss the phenomenon of superconductivity and its types. Show that in superconducting state current can flow in the superconducting specimen even if the electric field is removed (i.e. persistent current).

(CO2, CO3)

(b) Write all four Maxwell's equations in differential form. Derive the first, second and third Maxwell's equations and discuss their significance.

(CO2, CO6)

(c) Discuss the change in band gap if the particle size is reduced to nanometer.

(CO2, CO6)

Roll No.

TCH-101

**B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

ENGINEERING CHEMISTRY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is meant by bond order ? Explain its significance and calculate the bond order of He_2^+ , HF. (CO1)
- (b) Account for the following : (CO1)
 - (i) Density of ice is less than water
 - (ii) Para-nitrophenol is more soluble than o- nitrophenol
 - (iii) Boiling point trend in hydrides of 14 group (H_2O , H_2S , H_2Se , H_2Te)
 - (iv) O_2 has lower dissociation energy than o_2^+
- (c) Give the differences between UV-Visible and IR Spectroscopy. (CO1)

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2. (a) Describe the demineralization process for water softening in detail. Also give the chemical reactions involved in the process. (CO2)

(b) Calculate the amount of lime and soda required for soften 20,000 litres of water whose analysis is as follows :

Hardness in terms of $\text{Ca}(\text{HCO}_3)_2 = 40.5$ ppm; $\text{Mg}(\text{HCO}_3)_2 = 36.5$ ppm,
 $\text{CaSO}_4 = 34.0$ ppm; $\text{MgSO}_4 = 30.0$ ppm, $\text{CaCl}_2 = 27.75$ ppm;
 $\text{NaCl} = 25.0$ ppm; $\text{SiO}_2 = 40.9$ ppm (CO2)

(c) A exhausted zeolite softener was regenerated by passing 150 litre of NaCl containg 100g/l of NaCl. How many liters of water having hardness of 500 ppm have been softened by this zeolite bed ? (CO2)

3. (a) (i) Enumerate the difference between functionalty and tacticity.

(ii) Calculate the degree of polymerization of vinyl chloride, if the number average weight of polymer is 1.31×10^5 g/mol. (CO3)

(b) Give the preparation, monomer unit and repeating unit for the following :

(CO3)

(i) Polymer which is formed by cross condensation process, thermosetting in nature and bad conductor of electricity

(ii) Polymer having amide linkage and formed by cross condensation process

(iii) Polymer used for making bullet proof jackets

(iv) Polymer used in textile industry

(c) What are Biodegradable polymers ? Give the preparation of any *two* biodegradable polymers and their applications. (CO3)

4. (a) Define calorific value of a fuel. How is the calorific value of a fuel determined bomb calorimeter ? Explain with the help of diagram and give the formula for determination of GCV and NCV of a fuel. (CO4)
- (b) What is the condition when GCV is equal to NCV ? Calculate GCV and NCV of a fuel sample from the following data :

Weight of fuel = 0.83 gm

Water equivalent mass of calorimeter = 3500 gm

Amount of water taken in calorimeter = 385 gm

Initial temp. = 25.5°C

Final temperature = 29.2°C

% of H = 0.7%

Given latent heat of condensation of steam is 587 cal/gm. (CO4)

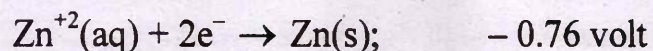
- (c) Explain the following : (CO4)

- (i) Composition of Biogas and ambient condition for the production of biogas
- (ii) Classification of fuels with example

5. (a) Define standard reduction electrode potential. Discuss electrochemical series and its applications. (CO5)

- (b) Justify the following : (CO5)

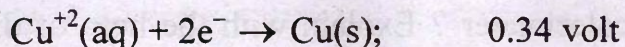
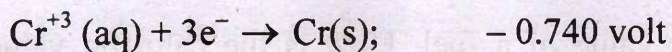
- (i) Can a solution of ZnSO_4 be stored in a vessel made up of copper.
- (ii) The standard reduction potential at 25°C for the following half-reaction is given below :



TMA-101
TEV-101
THU-101
THP-101

(4)

TCH-101



Which is the strongest reducing agent and why ?

(iii) Why blue colour of the solution gradually fades when CuSO_4 solution is electrolysed using platinum electrodes ?

(iv) Zn reacts with acid to give hydrogen gas where's Cu does not.

(c) For the cell ; (CO5)

$\text{Zn}(\text{s}) \mid \text{Zn}^{2+} (10^{-4} \text{ M}) \parallel \text{Mg}^{2+} (10^{-3} \text{ M}) \mid \text{Mg}(\text{s})$, the standard reduction potential for Zn and Mg electrodes are -0.764 V and -2.364 V .

(i) Identify the cathode and the anode.

(ii) Write the half cell reaction and overall cell reaction.

(iii) Calculate the EMF of cell and predict the spontaneity of cell reaction.